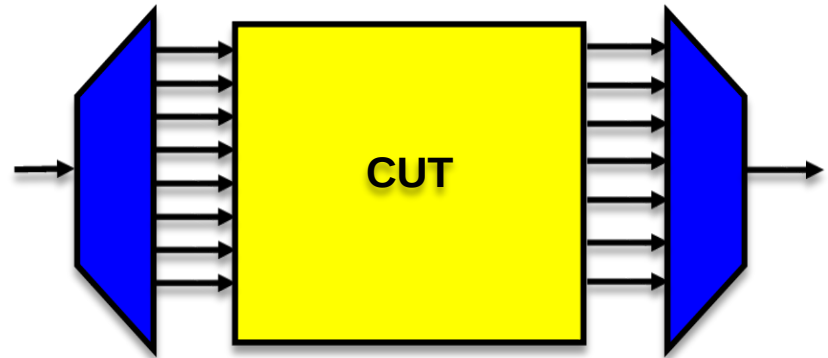


Test Compression

- Introduction
- Software Techniques
- Hardware Techniques
 - ♦ Test Stimulus Compression
 - ♦ Test Response Compaction(TRC)*
- Industry Practices
- Conclusion

*** NOTE**

Test stimulus **compression** is lossless
Test response **compaction** can be lossy



What is Good TRC?

1. High Compaction Ratio (CR)

$$CR = \frac{\text{Original Data Volume}}{\text{Compacted Data Volume}}$$

2. Low Aliasing

$$PAL = \frac{\text{number of faulty outputs that generate gold signature}}{\text{total number of faulty outputs}}$$

3. Tolerate/mask unknown (X) outputs

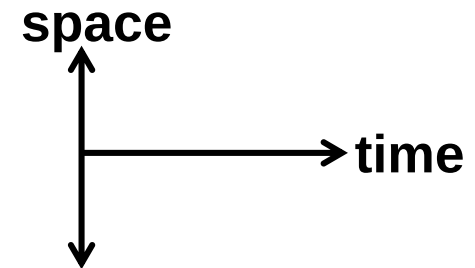
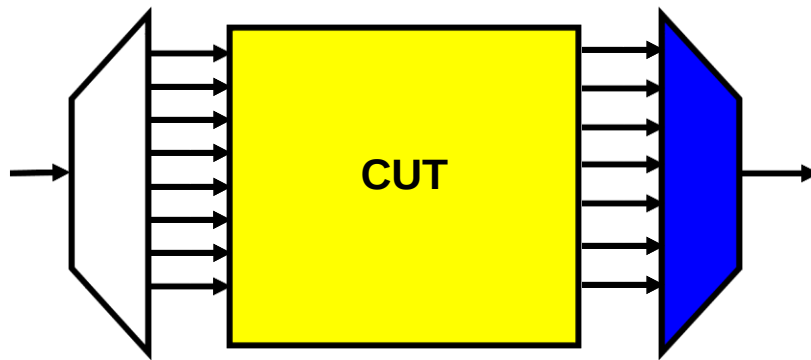
- ♦ Unknown outputs come from memory or non-scan flip-flop
- ♦ NOTE: this is different from **unspecified bit (X)** during ATPG



4. Diagnosis support (not in this lecture)

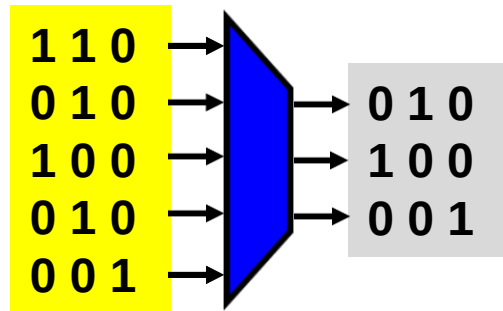
Compacted test responses of a fault is different from those of another fault

Test Response Compactor (TRC)



- Space compaction**

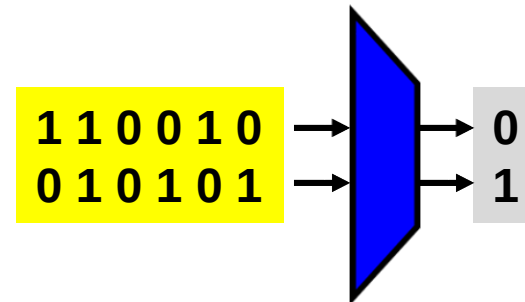
- ♦ reduces output pins



$$\text{Compaction Ratio} = \frac{5}{3}$$

- Time compaction**

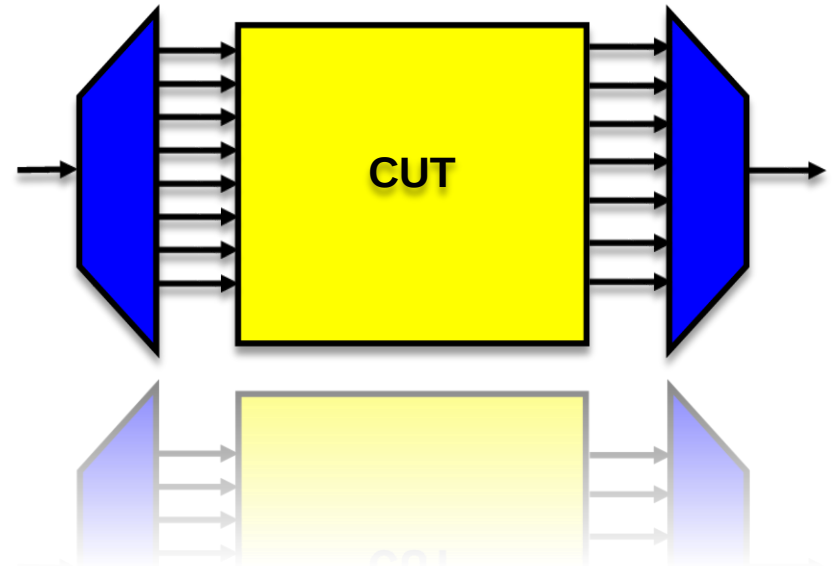
- ♦ reduces output length



$$\text{Compaction Ratio} = \frac{6}{1}$$

Test Compression

- Introduction
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 - X-compact
 - * Time Compaction
 - MISR
 - * Other X-handling techniques
 - X-blocking
 - X-masking
- Industry Practices
- Conclusion

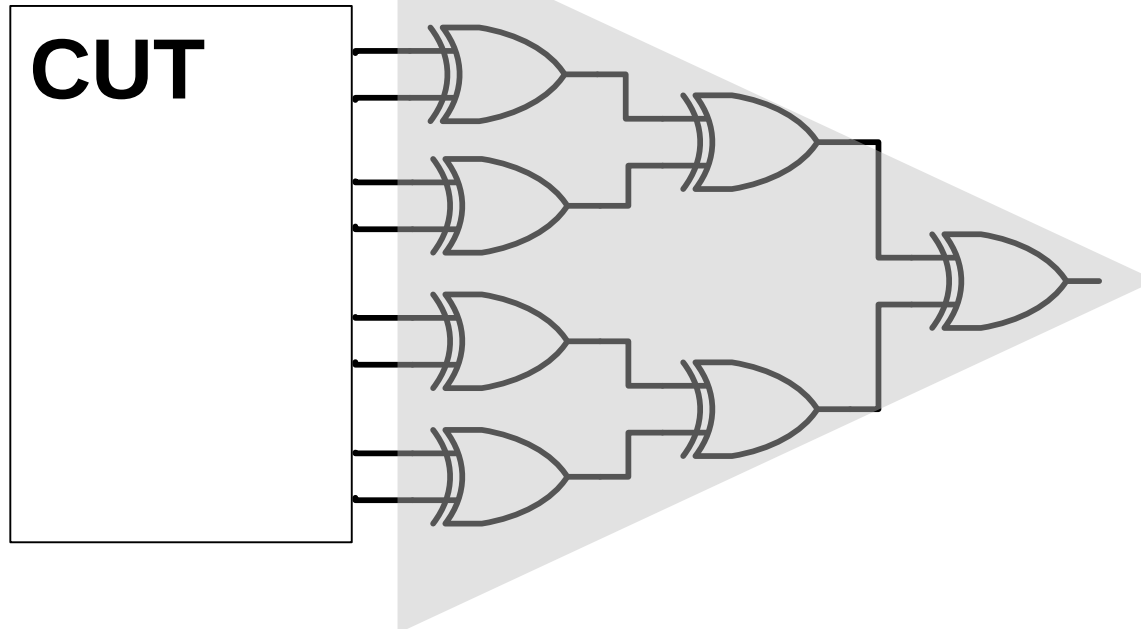


Single XOR-Tree

- 1. High CR
- 2. Bad PAL 
 - ♦ Detects **odd** number of errors, not even
- 3. What happens if X?

$$CR = \#CUT \text{ outputs}$$

$$PAL = \frac{1}{2}$$

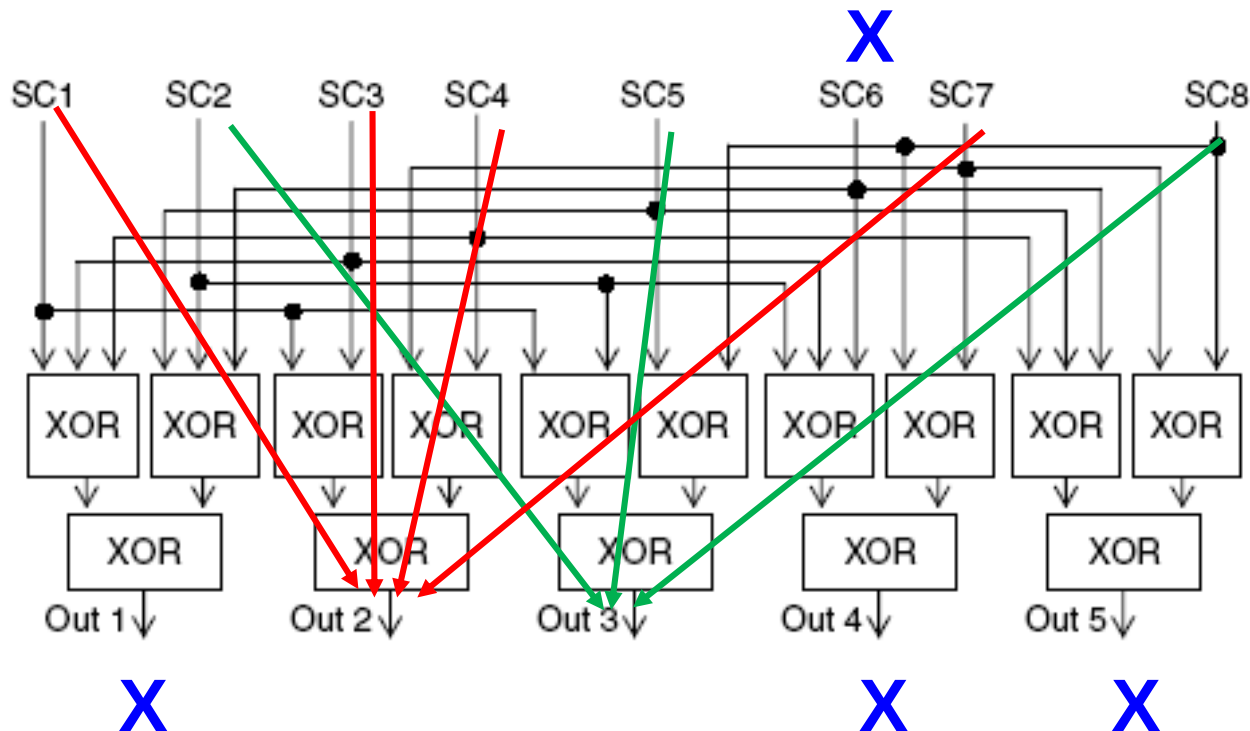


Single XOR Tree Can NOT Tolerate X

Idea: can we add more trees?

X-compact [Mitra 04]

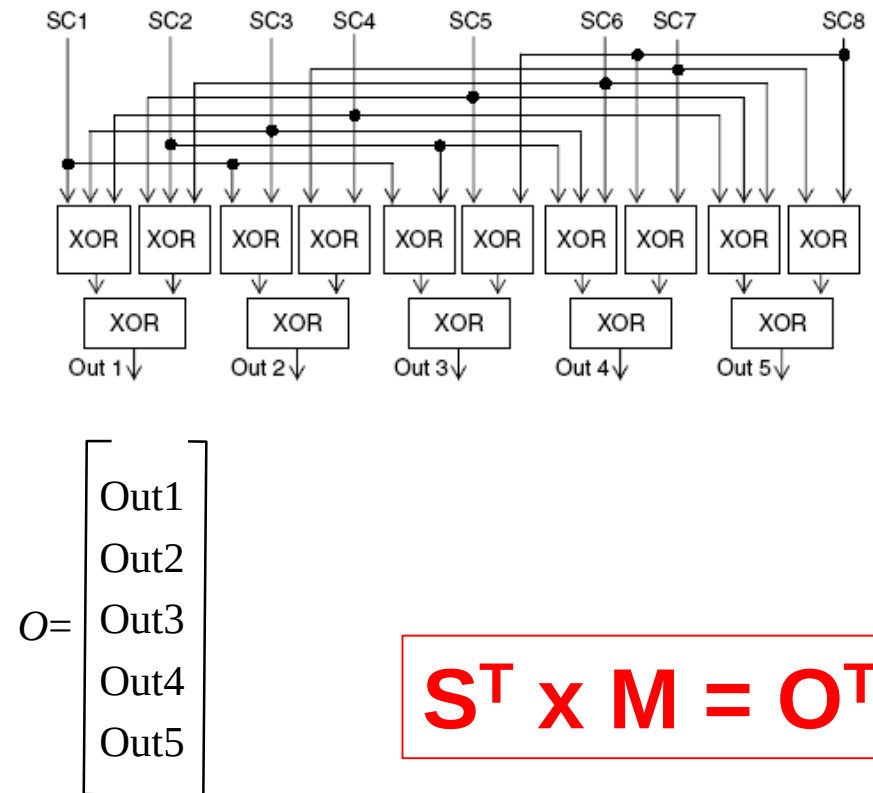
- Multiple XOR trees can detect errors in presence of X
- Example (WWW. Fig. 6.24)
 - ♦ Scan chain (SC) 6 produces unknown 'X'
 - ♦ The other 7 scan chains are not contaminated



X-compact Matrix, M

- Each row represents a scan chain
- Each column represents an compactor output
- $M_{i,j} = 1$ means j_{th} compactor outputs depends on i_{th} scan output

$$S = \begin{bmatrix} \text{SC1} \\ \text{SC2} \\ \text{SC3} \\ \text{SC4} \\ \text{SC5} \\ \text{SC6} \\ \text{SC7} \\ \text{SC8} \end{bmatrix} \quad M = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix}$$



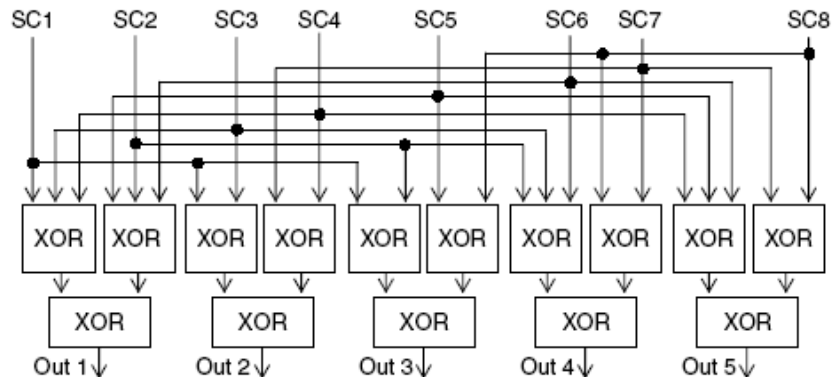
$$S^T \times M = O^T$$

X-compact Matrix, M

(WWW Theorem 6.4) Any 1, 2 or odd number of errors at same cycle are detected if every row in M has *distinct odd number of 1's*.

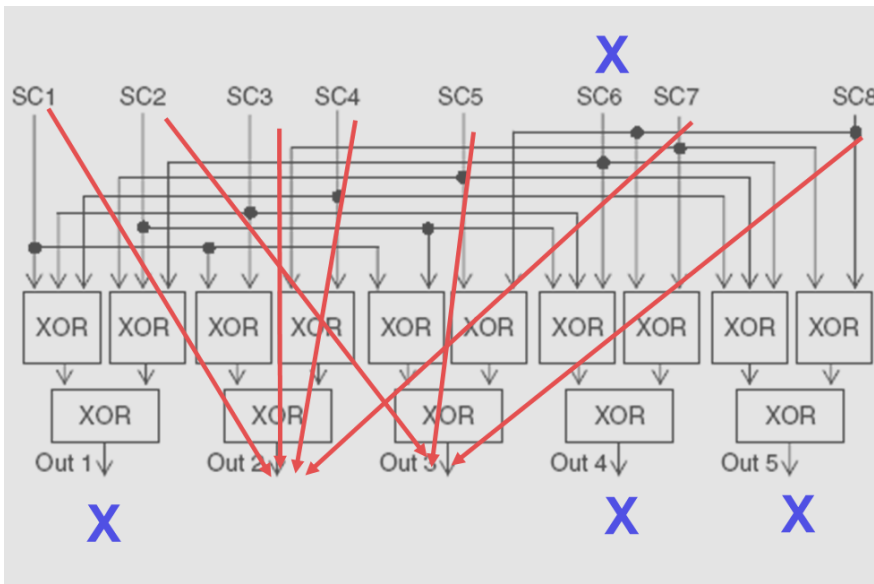
- 1) Single error is detected because no row is all zeros
- 2) Two errors are detected because adding any two rows produces non-zero results since no two rows are the same
- 3) Odd number of errors are detected because adding odd number of rows produces non-zero results (since all rows has odd 1's)

$$M = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix}$$



Theorem for X-tolerance

- X-compact guarantees to **detect 1 error** from any scan chain with **1 unknown (X)** from any other scan chain at same cycle
 - ♦ If and only if submatrix obtained by removing that row and columns having 1's
 - ♦ does not contain **a row of all 0s**
- Example: SC 6 produces X



$$M = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 \\ \rightarrow 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix}$$

What is CR?

- Every row in the X-compact matrix is *nonzero, distinct*
 - ♦ contains *odd number of 1's*

number of compactor outputs (#out)	max number of scan chains (#sc)	CR
5	$C_3^5=10$	2
6	$C_3^6=20$	3.3
7	$C_3^7=35$	5
8	$C_3^8=56$	7
9	$C_5^9=126$	14
10	$C_5^{10}=252$	25.2

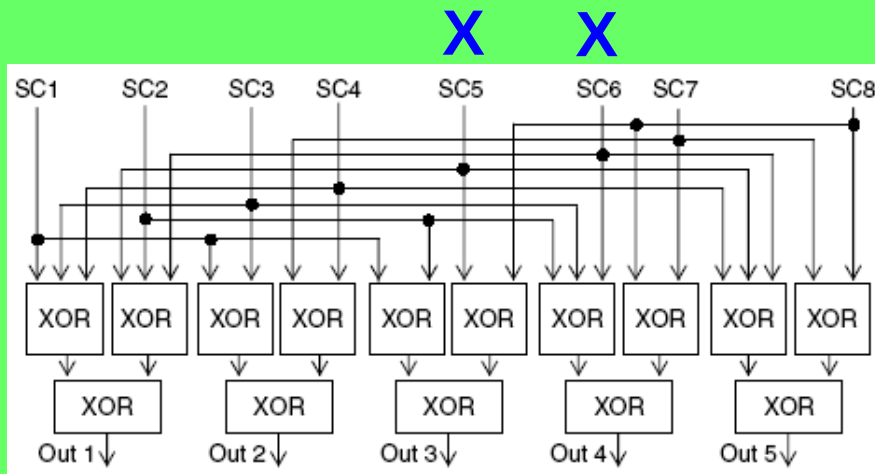
$$M = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \end{bmatrix} \left. \vphantom{\begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \end{bmatrix}} \right\} C_3^5 = 10$$

$$CR = \frac{\text{Original Data}}{\text{Compacted Data}} = \frac{\#SC}{\#Out}$$

QUIZ

Q: Which scan chain error we can NOT detect, when there are 2 X's from SC5 and SC6?

ANS:

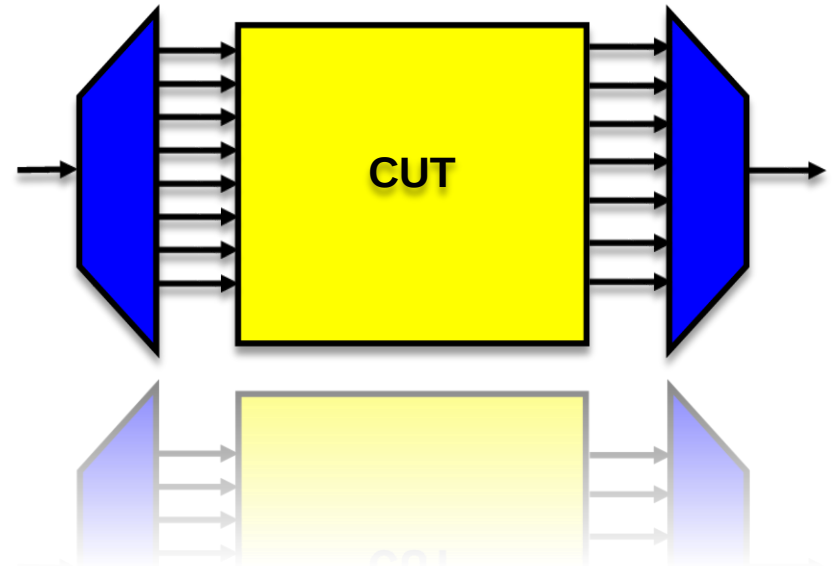


$$M = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix}$$

Cannot Tolerate Many X at Same Time

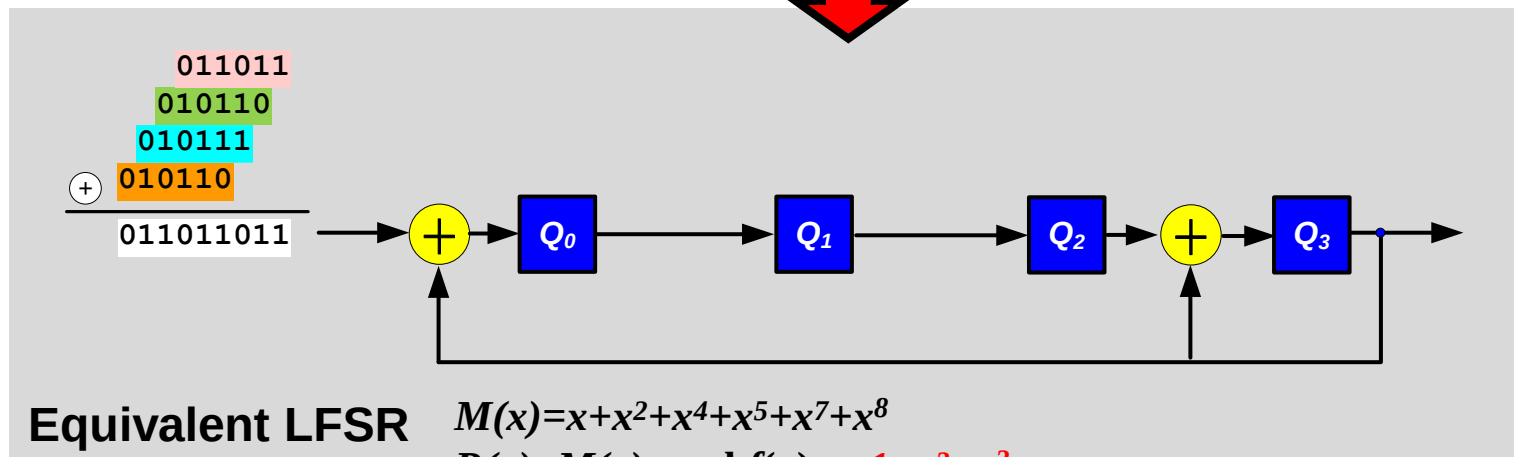
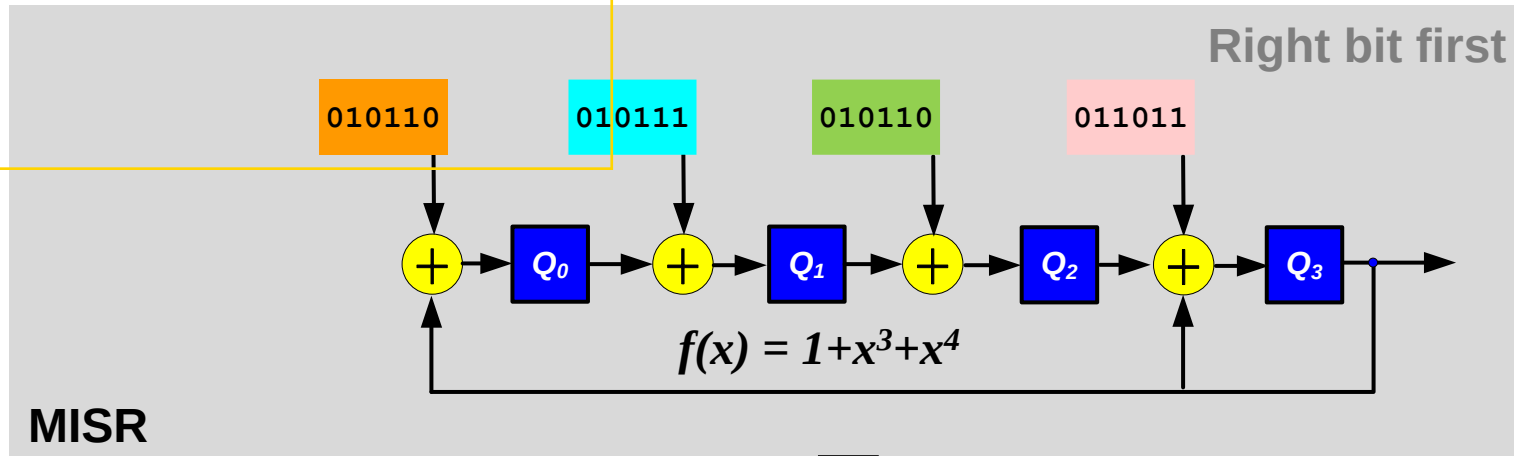
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Review: MISR (video 14.3)

- **MISR (multiple input signature register)** is similar to LFSR
 - ♦ except parallel inputs feed XOR between stages

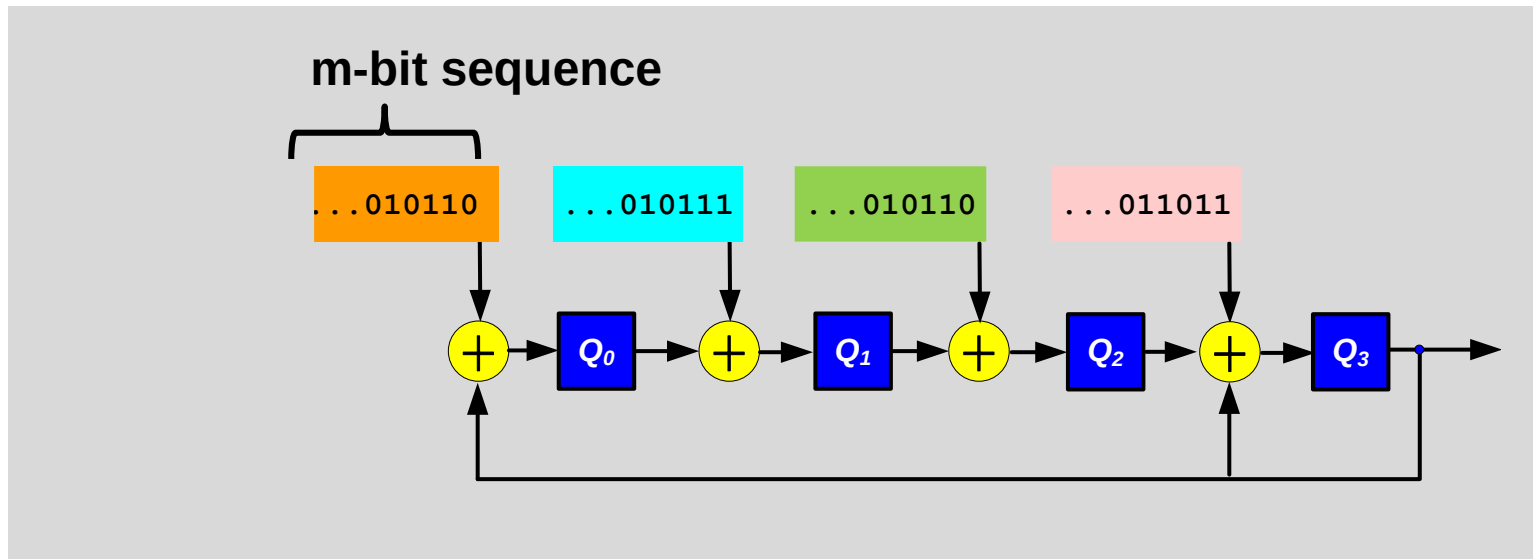


CR=? Aliasing=?

- MISR degree = N , input bit sequence length = m
 - ♦ Signature is N bits

$$CR = \frac{\text{Original Data}}{\text{Compacted Data}} = \frac{N \times m}{N} = m$$

$$PAL \approx 2^{-N} \quad (\text{see 14.3})$$

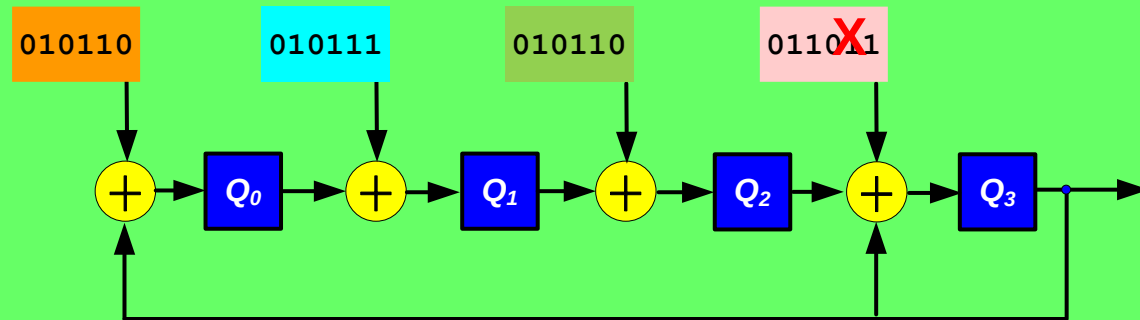


MISR has High CR and Low PAL

QUIZ

Q: What is signature if one bit is changed to X 'unknown' ?

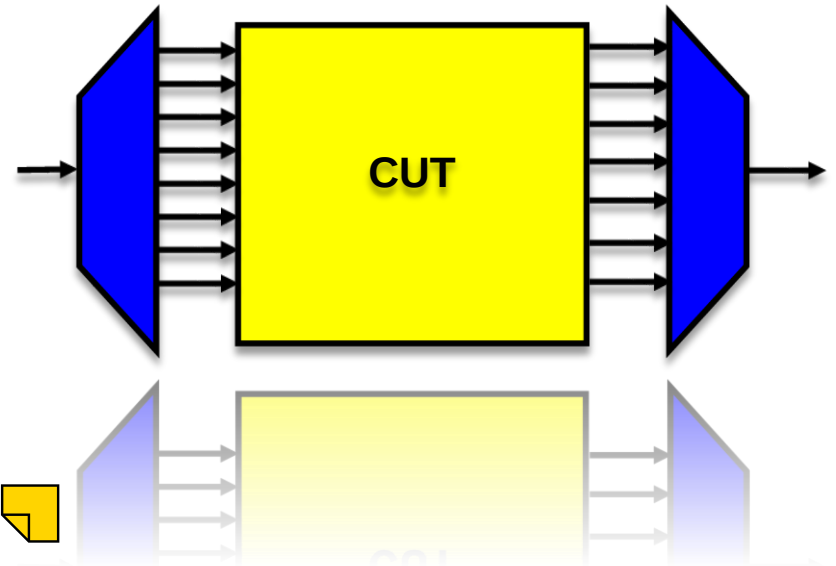
ANS: 



MISR is NOT X-tolerant

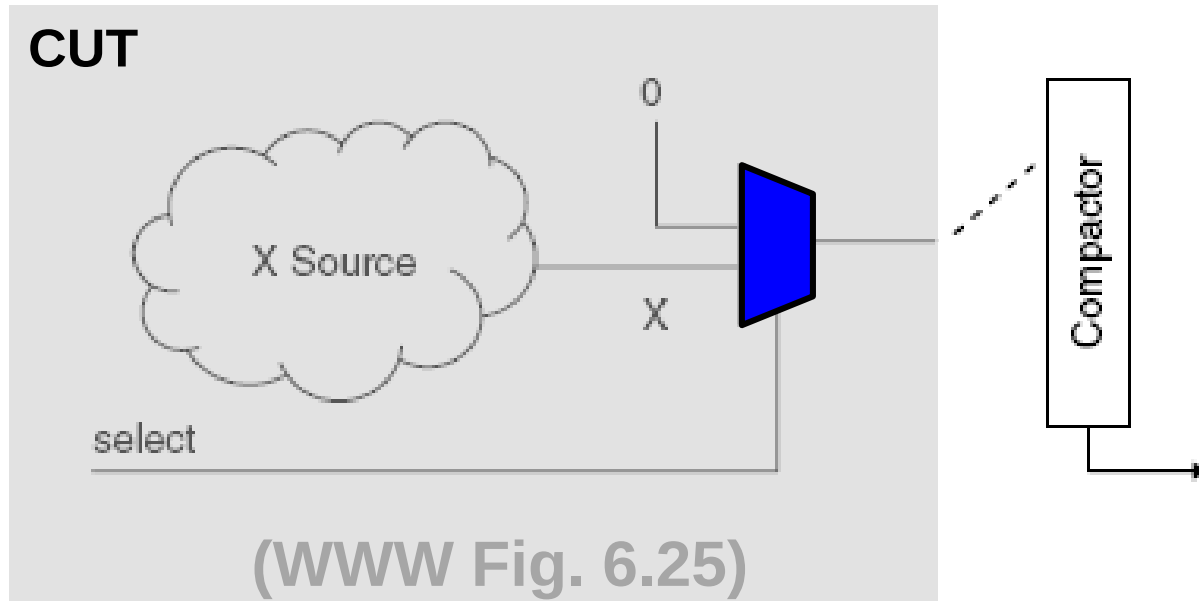
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X-blocking (or X-bounding)

- Add extra DFT inside CUT to block X before reaching compactor
 - ♦ Area overhead and extra delay
- X source can be
 - ♦ non-scan FF, memory, multi-cycle paths, false paths*...

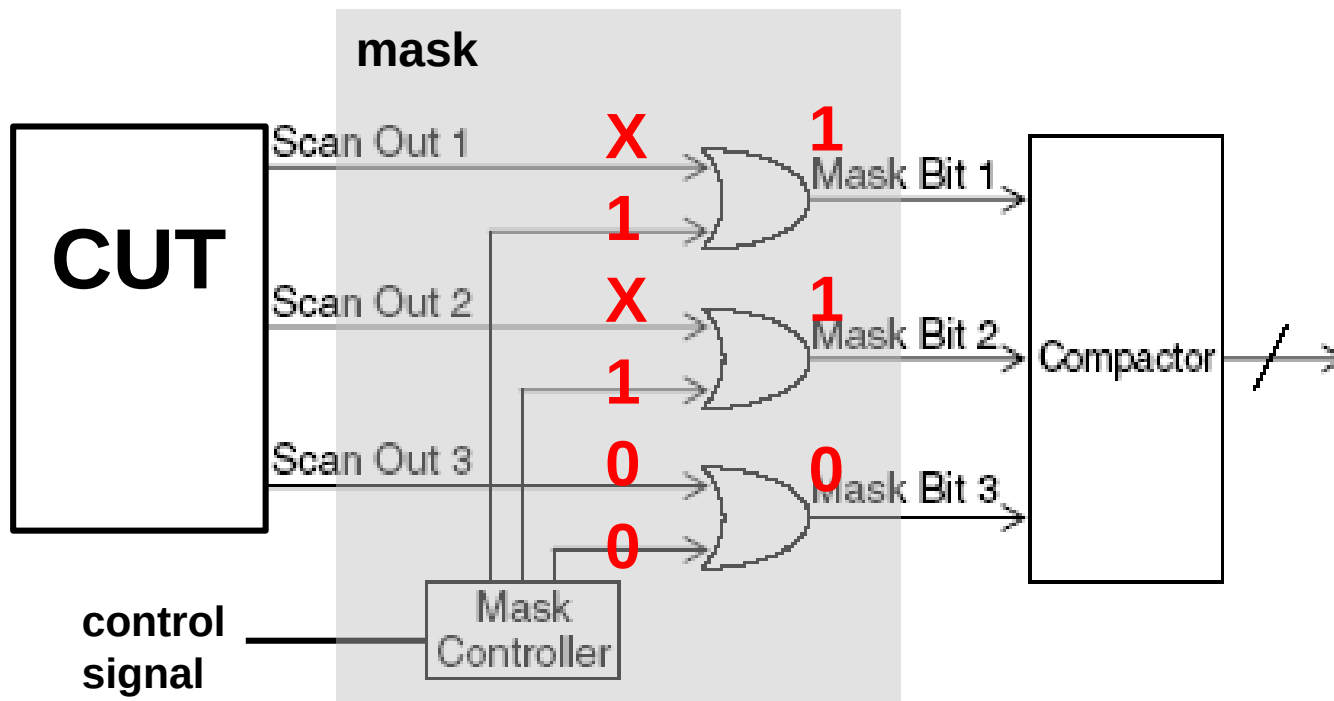


*multi-cycle paths needs more than 1 cycle to finish computation so test responses can be X

*false paths are not activated by normal operation so test responses can be X

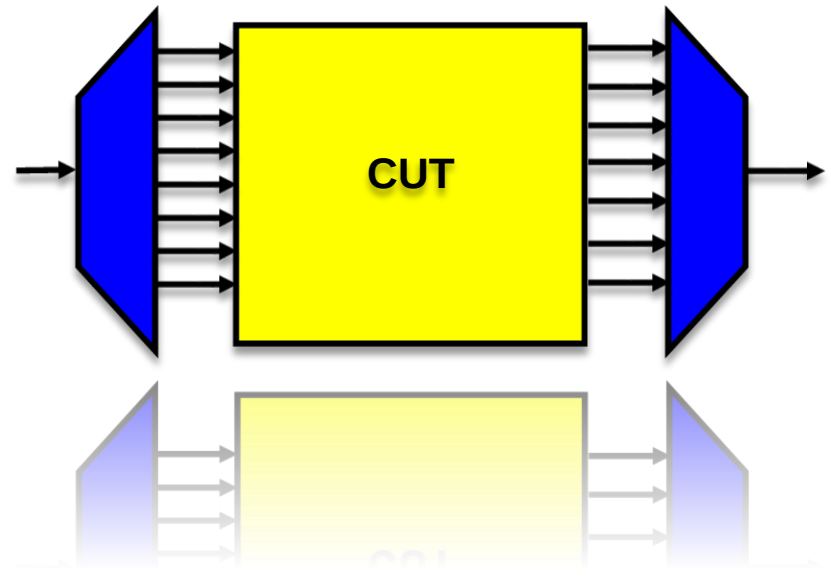
X-masking

- Add extra mask between CUT and compactor
- Example: mask outputs by OR gates
 - ♦ 1 = mask
 - ♦ 0 = pass through



Summary

- **Test Response Compaction**
 - ◆ Space Compaction
 - * XOR-tree, X-compact
 - ◆ Time Compaction
 - * MISR
 - High CR, Low PAL
 - Cannot tolerate X
 - ◆ X-bounding, X-masking
 - Can mask many X

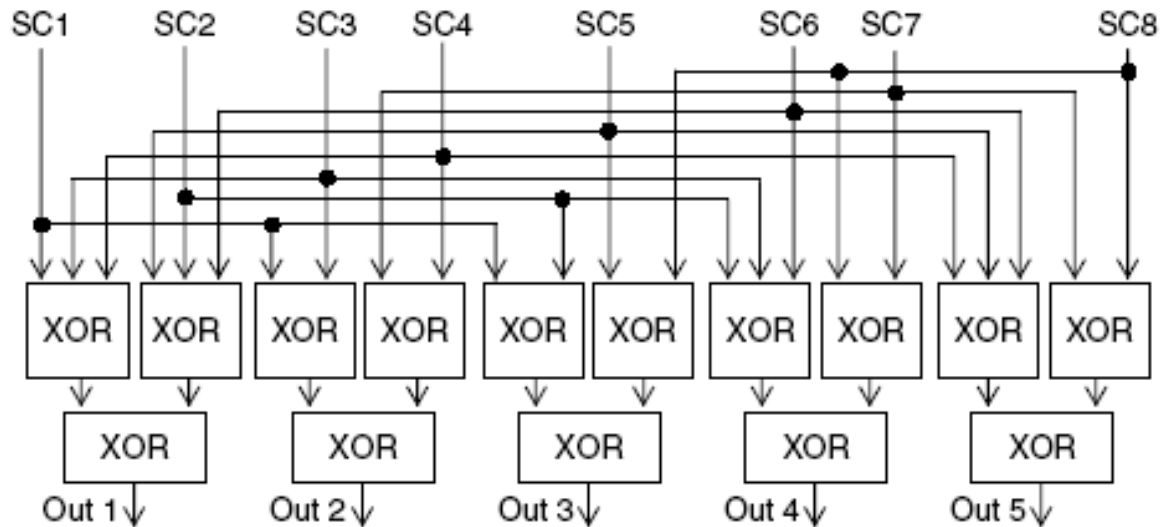


**X-Masking + MISR/XOR-tree
is Most Popular Solution**

FFT

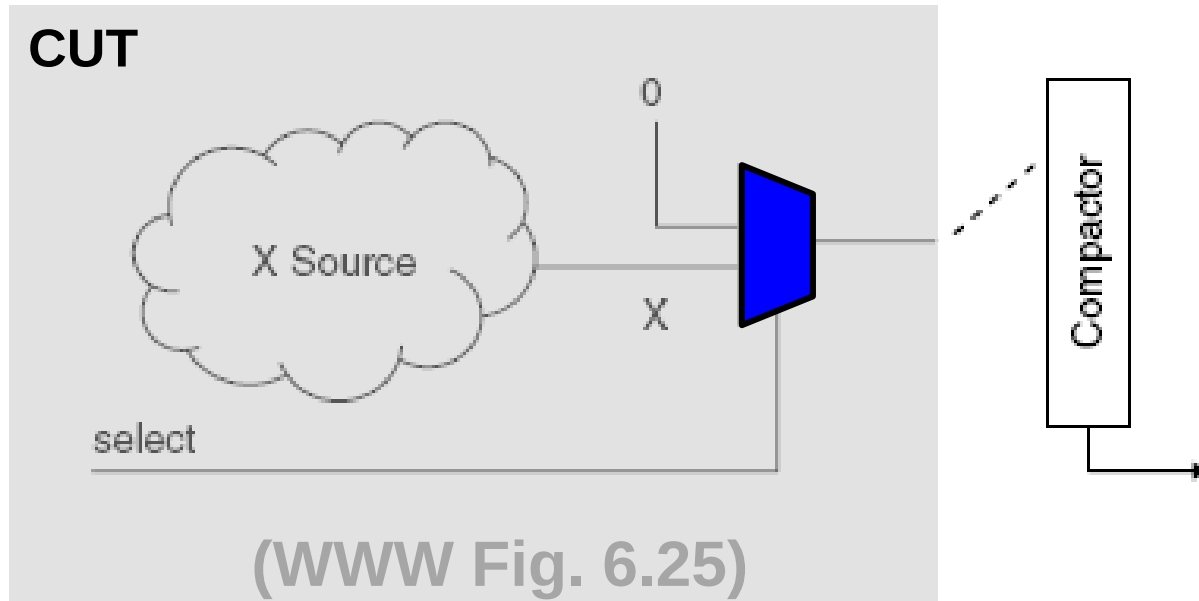
- What is Prob. of Aliasing for X-compactor?

(Theorem 6.4) Any **1, 2 or odd number** of errors at same scan-out cycle are detected if **every row in M has distinct odd number of 1's.**



FFT

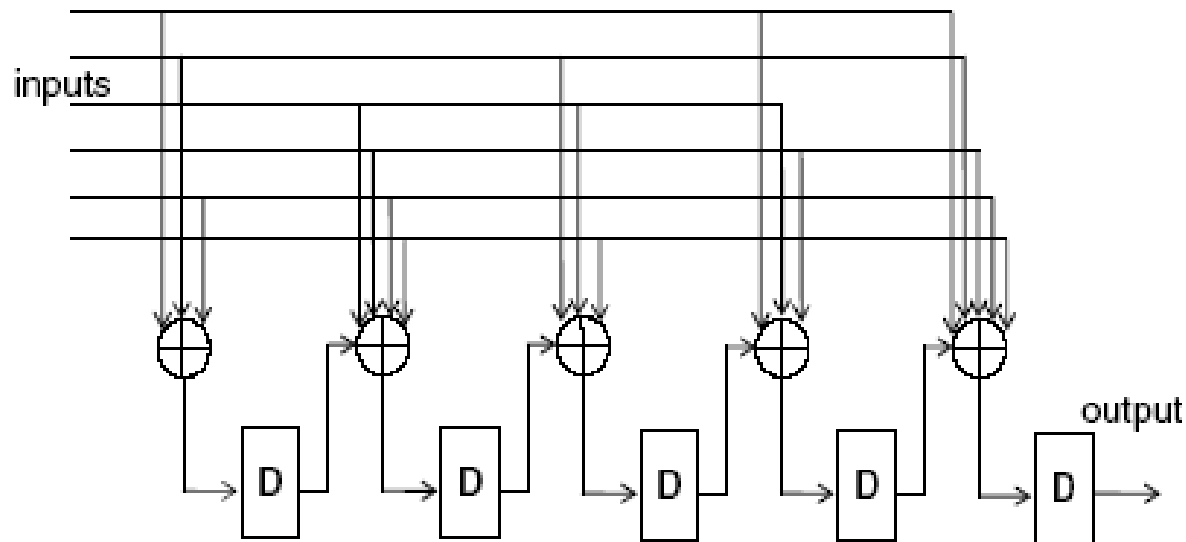
- X source can be multi-cycle paths, false paths
- Q: why multi-cycle paths generate X in test mode?
- Q: why false paths generate X in test mode?



*multi-cycle paths needs more than 1 cycle to finish computation so test responses can be X
*false paths are not activated by normal operation so test responses can be X

FFT

- This is a **hybrid space-time compactor**
- Q: What are advantages and disadvantages ?



FFT

- Q: In X-compact matrix, why cannot we have **even number of 1's** in each row?

$$\begin{bmatrix} 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Theorems for Error Detection

- (WWW Theorem 6.3)
 - ◆ If only a single scan chain produces an error at any scan-out cycle (scan slice), the X-compactor is guaranteed to produce errors if and only if *no row of the X-compact matrix contains all 0's.*
- (WWW Theorem 6.4)
 - ◆ Errors from any **one , two or an odd number** of scan chains at the same scan-out cycle are guaranteed to be detected
 - * if **every row in the X-compact matrix is *nonzero, distinct and contains an odd number of 1's.***

How to Design X-compactor?

- (WWW Theorem 6.4)
 - ♦ Errors from any **one , two or an odd number** of scan chains at the same scan-out cycle are guaranteed to be detected
 - ♦ if **every row in the X-compact matrix is *nonzero, distinct and contains an odd number of 1's***.

max number of scan chains (#sc)	number of compactor outputs (#out)
$C^5_3=10$	5
$C^6_3=20$	6
$C^7_3=35$	7
$C^8_3=56$	8
$C^9_5=126$	9
$C^{10}_5=252$	10

$$M = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix}$$

$$CR = \frac{\text{Original Data}}{\text{Compressed Data}} = \frac{\#SC}{\#Out}$$

Time v.s. Space Compaction

- **D**: original test responses
- **C**: compacted test responses
- Compactor converts D matrix ($m \times n$) to C matrix ($p \times q$)
 - ♦ Column index referred to as time dimension
 - ♦ Row index referred to as space dimension
- Space compression: $p < m, q = n$
- Time compression: $q < n$

$$C = \Phi(D)$$

